

VRAAG / QUESTION 1

'n Deeltjie beweeg in 'n reguitlyn in die x-as rigting en die grootte van sy snelheid is v .
 A particle is moving in a straight line in the x-direction and the magnitude of its velocity is v .

- (a) Definieer sy snelheid v en gee die eenheid. / Define its velocity v and give its unit.

Tempo waarlangs hy teen die x-as af beweeg
 in m/s rate of displacement in x-direction

- (b) Met $v = v_0 + at$, toon deur middel van integrasie aan dat vir 'n verplasing tussen x_0 en x in 'n tyd t ($t_0 = 0$) is $x - x_0 = v_0 t + \frac{1}{2} at^2$.

With $v = v_0 + at$, show by means of integration that for a displacement between x_0 and x in a time t ($t_0 = 0$) is $x - x_0 = v_0 t + \frac{1}{2} at^2$.

$$dv = \frac{dx}{dt} \quad a = \frac{dv}{dt} \quad dt = \frac{dx}{v} \quad \therefore dx = v dt$$

$$\int_{x_0}^x dx = \int_{t_0}^t v dt \rightarrow x - x_0 = \int_0^t (v_0 + at) dt$$

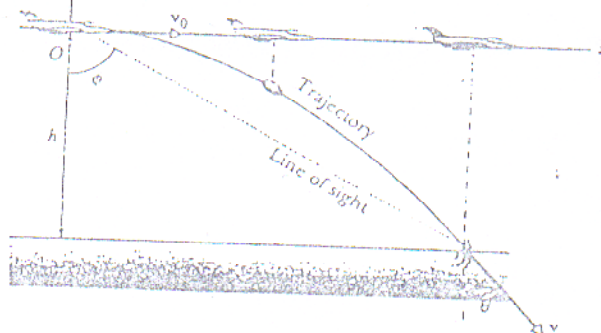
$$x - x_0 = (v_0 t + \frac{1}{2} at^2) \Big|_0^t$$

$$x - x_0 = (v_0 t + \frac{1}{2} at^2 - 0)$$

$$x - x_0 = v_0 t + \frac{1}{2} at^2 \quad (3)$$

- (c) 'n Vliegtuig vlieg teen $v_0 = 430$ km/h op 'n hoogte van 1200 m bo die see en laat 'n reddingspak val vir 'n drenkeling in die see.

An aeroplane has a speed $v_0 = 430$ km/h and is flying at a height of 1200 m above the sea. It drops a parcel to a person in the sea.



- (i) Toon aan dat die tyd wat die reddingspak deur die lug trek is $t = 15,65$ s.
 Show that the time of flight for the parcel is $t = 15,65$ s.

$$y - y_0 = v_0 t + \frac{1}{2} at^2$$

$$1200 = (119,44) t + \frac{1}{2} (-9,8) t^2$$

$$0 = 4,9 t^2 + 119,44 t - 1200$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad t = \frac{-119,44 \pm \sqrt{3778,97}}{9,8}$$

$$0 = 4,9 t^2 - 1200$$

$$t^2 = 244,89$$

Vertical comp. of motion:

$$y - y_0 = v_{y0} t + \frac{1}{2} a_y t^2 \quad \therefore t = 15,65$$

$$-1200 - 0 = 0 t + \frac{1}{2} (-9,80) t^2$$

$$1200 = \frac{9,80}{2} t^2$$

$$t^2 = 244,9$$

$$t = 15,65$$

Vraag / Question 1(c)(ii)(vervolg)(continue) / ...